

HOMEWORK 4
ELEMENTARY LINEAR ALGEBRA
(MATH 2250-03), SPRING, 2018

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SECTION 2.2 - THE INVERSE OF A MATRIX

Let $A = \begin{bmatrix} 3 & 2 \\ 7 & 4 \end{bmatrix}$, let $B = \begin{bmatrix} 3 & -4 \\ 7 & -8 \end{bmatrix}$, and let $C = \begin{bmatrix} 1 & 0 & -2 \\ -3 & 1 & 4 \\ 2 & -3 & 4 \end{bmatrix}$.

1. Compute $(AB)^{-1}$.

We aim to compute $\{AB\}^{-1}$. To that end, let us recall that if A and B are invertible matrices, then

$$(AB)^{-1} = B^{-1}A^{-1}.$$

So, if A and B are invertible, then we may calculate $(AB)^{-1}$ using A^{-1} and B^{-1} . Let us now recall that a 2×2 matrix M of the form $M = \begin{bmatrix} a & b \\ c & d \end{bmatrix}$ satisfying $ad - bc \neq 0$ is invertible and $M^{-1} = \frac{1}{ad - bc} \begin{bmatrix} d & -b \\ -c & a \end{bmatrix}$. So, observe that

$$(3)(4) - (2)(7) = 12 - 14 = -2 \neq 0$$

and

$$(3)(-8) - (-4)(7) = -24 + 28 = 4 \neq 0.$$

Thus, we may conclude both A and B are invertible, and we may compute

$$A^{-1} = -\frac{1}{2} \begin{bmatrix} 4 & -2 \\ -7 & 3 \end{bmatrix}, \quad B^{-1} = \frac{1}{4} \begin{bmatrix} -8 & 4 \\ -7 & 3 \end{bmatrix}.$$

So, using our initial observation that $(AB)^{-1} = B^{-1}A^{-1}$, we reach our final answer by computing

$$\begin{aligned} (AB)^{-1} &= B^{-1}A^{-1} \\ &= \left(\frac{1}{4} \begin{bmatrix} -8 & 4 \\ -7 & 3 \end{bmatrix} \right) \left(-\frac{1}{2} \begin{bmatrix} 4 & -2 \\ -7 & 3 \end{bmatrix} \right) \\ &= -\frac{1}{8} \begin{bmatrix} -8 & 4 \\ -7 & 3 \end{bmatrix} \begin{bmatrix} 4 & -2 \\ -7 & 3 \end{bmatrix} \\ &= -\frac{1}{8} \begin{bmatrix} -60 & 28 \\ -49 & 23 \end{bmatrix} \\ &= \begin{bmatrix} 15/2 & -7/2 \\ 49/8 & -23/8 \end{bmatrix}. \end{aligned}$$

2. Compute $(C^T)^{-1}$.

We aim to compute $(C^T)^{-1}$. To that end, let us recall that if C is an invertible matrix, then $(C^T)^{-1} = (C^{-1})^T$. So, if C is invertible, then we may calculate $(C^T)^{-1}$ using C^{-1} . Using the algorithm described in Section 2.2, we proceed as follows:

$$\begin{aligned}
 \left[\begin{array}{ccc|ccc} 1 & 0 & -2 & 1 & 0 & 0 \\ -3 & 1 & 4 & 0 & 1 & 0 \\ 2 & -3 & 4 & 0 & 0 & 1 \end{array} \right] &\Rightarrow \left[\begin{array}{ccc|ccc} 1 & 0 & -2 & 1 & 0 & 0 \\ 0 & 1 & -2 & 3 & 1 & 0 \\ 2 & -3 & 4 & 0 & 0 & 1 \end{array} \right] \text{ (Replace } R_2 \text{ with } R_2 + 3R_1) \\
 &\Rightarrow \left[\begin{array}{ccc|ccc} 1 & 0 & -2 & 1 & 0 & 0 \\ 0 & 1 & -2 & 3 & 1 & 0 \\ 0 & -3 & 8 & -2 & 0 & 1 \end{array} \right] \text{ (Replace } R_3 \text{ with } R_3 - 2R_1) \\
 &\Rightarrow \left[\begin{array}{ccc|ccc} 1 & 0 & -2 & 1 & 0 & 0 \\ 0 & 1 & -2 & 3 & 1 & 0 \\ 0 & 0 & 2 & 7 & 3 & 1 \end{array} \right] \text{ (Replace } R_3 \text{ with } R_3 + 3R_2) \\
 &\Rightarrow \left[\begin{array}{ccc|ccc} 1 & 0 & -2 & 1 & 0 & 0 \\ 0 & 1 & 0 & 10 & 4 & 1 \\ 0 & 0 & 2 & 7 & 3 & 1 \end{array} \right] \text{ (Replace } R_2 \text{ with } R_2 + R_3) \\
 &\Rightarrow \left[\begin{array}{ccc|ccc} 1 & 0 & 0 & 8 & 3 & 1 \\ 0 & 1 & 0 & 10 & 4 & 1 \\ 0 & 0 & 2 & 7 & 3 & 1 \end{array} \right] \text{ (Replace } R_1 \text{ with } R_1 + R_3) \\
 &\Rightarrow \left[\begin{array}{ccc|ccc} 1 & 0 & 0 & 8 & 3 & 1 \\ 0 & 1 & 0 & 10 & 4 & 1 \\ 0 & 0 & 1 & 7/2 & 3/2 & 1/2 \end{array} \right] \text{ (Replace } R_3 \text{ with } (1/2)R_3)
 \end{aligned}$$

The inverse of C is the right half of this augmented matrix, so

$$C^{-1} = \begin{bmatrix} 8 & 3 & 1 \\ 10 & 4 & 1 \\ 7/2 & 3/2 & 1/2 \end{bmatrix}.$$

Hence, we may conclude by computing

$$\begin{aligned}
 (C^T)^{-1} &= (C^{-1})^T \\
 &= \begin{bmatrix} 8 & 10 & 7/2 \\ 3 & 4 & 3/2 \\ 1 & 1 & 1/2 \end{bmatrix}^T \\
 &= \begin{bmatrix} 8 & 3 & 1 \\ 10 & 4 & 1 \\ 7/2 & 3/2 & 1/2 \end{bmatrix}.
 \end{aligned}$$

SECTION 2.3 - CHARACTERIZATIONS OF INVERTIBLE MATRICES

3. Let A and B be $n \times n$ matrices and suppose $BA = I$. Show that A is invertible and that $B = A^{-1}$.

Let A and B be $n \times n$ matrices and suppose $BA = I$. We desire to show that A is invertible and that $B = A^{-1}$. To that end, recall that the Invertible Matrix Theorem (Theorem 8 in section 2.3 of the text) states, among other things, that a square matrix A is invertible if and only if the homogeneous system $Ax = 0$ has only the trivial solution $x = 0$. So, observe that if $x \in \mathbb{R}^n$ is any vector such that $Ax = 0$, then

$$Ax = 0$$

$$B(Ax) = B0$$

$$(BA)x = 0$$

$$Ix = 0$$

$$x = 0.$$

So, if $x \in \mathbb{R}^n$ is such that $Ax = 0$, then it must be the case that $x = 0$. That is, the homogeneous system $Ax = 0$ has only the trivial solution. So, it follows from the Invertible Matrix Theorem that A is invertible. Therefore, we may conclude by making the following calculation:

$$BA = I$$

$$BAA^{-1} = IA^{-1}$$

$$B = A^{-1}.$$

4. Let A and B be $n \times n$ matrices and suppose AB is invertible. Show that A is invertible and give a specific formula for the inverse of A in terms of $(AB)^{-1}$ and B . (HINT: Use what you showed in problem 3.)

Supposing A and B be $n \times n$ matrices such that AB is invertible, we seek to show that A is invertible. To do so, let us note that

$$(AB)(AB)^{-1} = I.$$

So, using the associative property of matrix multiplication, we then have

$$A(B(AB)^{-1}) = I.$$

Hence, the matrix $C = B(AB)^{-1}$ satisfies $AC = I$. Using the result of problem 4, it then follows that C is invertible and $A = C^{-1}$. So, noting that $AC = CC^{-1} = I$ and $CA = CC^{-1} = I$, it is clear that A is invertible. Moreover, recalling that $(C^{-1})^{-1} = C$, we have

$$A^{-1} = (C^{-1})^{-1} = B(AB)^{-1}.$$

Side note: while the above is enough to finish the problem, it turns out that our answer can actually be put into an *even simpler* form. If you want a fun challenge, try showing that $A^{-1} = B$.

SECTION 2.4 - PARTITIONED MATRICES

5. Suppose A , B , and C are $n \times n$ matrices, and let X , Y , and Z be $n \times n$ matrices such that

$$\begin{bmatrix} X & 0 \\ Y & Z \end{bmatrix} \begin{bmatrix} A & 0 \\ B & C \end{bmatrix} = \begin{bmatrix} I & 0 \\ 0 & I \end{bmatrix}.$$

Find formulas for X , Y , and Z in terms of A , B , and C .

(HINT: Use what you showed in problem 3.)

As A , B , C , X , Y , and Z are all $n \times n$ matrices, we may compute the product of the block matrices $\begin{bmatrix} X & 0 \\ Y & Z \end{bmatrix}$ and $\begin{bmatrix} A & 0 \\ B & C \end{bmatrix}$ as if they were regular 2×2 matrices, provided we take care to account for the order of multiplication. That is,

$$\begin{aligned} \begin{bmatrix} X & 0 \\ Y & Z \end{bmatrix} \begin{bmatrix} A & 0 \\ B & C \end{bmatrix} &= \begin{bmatrix} (XA + 0B) & (X0 + 0C) \\ (YA + ZB) & (Y0 + ZC) \end{bmatrix} \\ &= \begin{bmatrix} XA & 0 \\ (YA + ZB) & ZC \end{bmatrix}. \end{aligned}$$

So, if $\begin{bmatrix} X & 0 \\ Y & Z \end{bmatrix} \begin{bmatrix} A & 0 \\ B & C \end{bmatrix} = \begin{bmatrix} I & 0 \\ 0 & I \end{bmatrix}$, then it must be the case that

$$\begin{bmatrix} XA & 0 \\ (YA + ZB) & ZC \end{bmatrix} = \begin{bmatrix} I & 0 \\ 0 & I \end{bmatrix}.$$

That is, X , Y , and Z must be such that

$$XA = I, \quad YA + ZB = 0, \quad \text{and} \quad ZC = I.$$

Using $XA = I$ and $ZC = I$ together with the result of problem 3, it follows that A and C are invertible and that $X = A^{-1}$ and $Z = C^{-1}$. Using this, we then have that

$$YA + ZB = YA + C^{-1}B.$$

So, by the invertibility of A , the equality $YA + ZB = 0$ becomes

$$\begin{aligned} YA + ZB &= 0 \\ YA + C^{-1}B &= 0 \\ YA &= -C^{-1}B \\ Y &= -C^{-1}BA^{-1}. \end{aligned}$$

Therefore, we may conclude that if A , B , and C are $n \times n$ matrices such that

$$\begin{bmatrix} X & 0 \\ Y & Z \end{bmatrix} \begin{bmatrix} A & 0 \\ B & C \end{bmatrix} = \begin{bmatrix} I & 0 \\ 0 & I \end{bmatrix},$$

then $X = A^{-1}$, $Y = -C^{-1}BA^{-1}$, and $Z = C^{-1}$.

SECTION 2.5 - MATRIX FACTORIZATIONS

6. Let $A = \begin{bmatrix} 4 & 3 & -5 \\ -4 & -5 & 7 \\ 8 & 6 & -8 \end{bmatrix}$ and let $b = \begin{bmatrix} 2 \\ -4 \\ 6 \end{bmatrix}$. Solve the equation $Ax = b$ by using the following LU factorization of A:

$$A = \begin{bmatrix} 1 & 0 & 0 \\ -1 & 1 & 0 \\ 2 & 0 & 1 \end{bmatrix} \begin{bmatrix} 4 & 3 & -5 \\ 0 & -2 & 2 \\ 0 & 0 & 2 \end{bmatrix}$$

We seek to solve the equation $Ax = b$ using the LU factorization of A given above. To do so, note that if $A = LU$ for a lower triangular matrix L and an upper triangular matrix U, then $Ax = b$ is equivalent to $LUx = b$. So, if we can solve $Ly = b$, then we need only solve $Ux = y$ to solve $LUx = b$ (which, in turn, provides a solution to $Ax = b$). That is, if $Ux = y$ and $Ly = b$, then

$$Ax = LUx = L(Ux) = Ly = b.$$

So, we first solve

$$\begin{aligned} & Ly = b \\ & \begin{bmatrix} 1 & 0 & 0 \\ -1 & 1 & 0 \\ 2 & 0 & 1 \end{bmatrix} \begin{bmatrix} y_1 \\ y_2 \\ y_3 \end{bmatrix} = \begin{bmatrix} 2 \\ -4 \\ 6 \end{bmatrix} \\ & \begin{bmatrix} y_1 \\ -y_1 + y_2 \\ 2y_1 + y_3 \end{bmatrix} = \begin{bmatrix} 2 \\ -4 \\ 6 \end{bmatrix}. \end{aligned}$$

Back substituting, this produces

$$y_1 = 2, \quad y_2 = -4 + y_1 = -2, \quad y_3 = 6 - 2y_1 = 2.$$

So, we have found a solution to $Ly = b$ as desired. Following this, we then solve

$$\begin{aligned} & Ux = y \\ & \begin{bmatrix} 4 & 3 & -5 \\ 0 & -2 & 2 \\ 0 & 0 & 2 \end{bmatrix} \begin{bmatrix} x_1 \\ x_2 \\ x_3 \end{bmatrix} = \begin{bmatrix} 2 \\ -2 \\ 2 \end{bmatrix} \\ & \begin{bmatrix} 4x_1 + 3x_2 - 5x_3 \\ -2x_2 + 2x_3 \\ 2x_3 \end{bmatrix} = \begin{bmatrix} 2 \\ -2 \\ 2 \end{bmatrix}. \end{aligned}$$

Back substituting, this produces

$$x_3 = 1, \quad x_2 = (-1/2)(-2 - 2x_3) = 2, \quad x_1 = (1/4)(2 - 3x_2 + 5x_3) = 1/4.$$

So, we have found a solution to $Ux = y$ as desired, and by the reasoning given above, we may

conclude that $x = \begin{bmatrix} 1/4 \\ 2 \\ 1 \end{bmatrix}$ is a solution to $Ax = b$